

#4 Korean GD HLA / Pan-Asian

*"Pan-Asian HLA architecture of Graves' disease – DPB1*05:01 / B*46:01 risk continuum and population-specific haplotype structure" – Paper 4* 는 *v8 Pillar I forest* 직속 후신이며, 4/4 entry condition 충족 시 unlock 되는 *backlog* 다. 현재 *v8 R1/D3/F1/M9 contents* 는 *parking material* 로 보존되어 있다.

Status: backlog

Execution: 4/4 gated

Substrate: Korean PTC pool n=874

Anchor: v19_paper4_GD_backlog

Renumbered: Paper 3 → Paper 4 · 2026-05-04

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I. 한 줄 결론

Risk continuum: DPB1*05:01 carrier **53.2% Korean PTC > 44.0% Han Chinese GD > 31.3% Han Chinese ctrl.** Pooled OR 2.16 [1.65–2.83] random-effects DerSimonian-Laird. Korean PTC 가 Han Chinese GD case 보다도 더 risky 한 HLA 구성. **Population-specific autoimmune-thyroid susceptibility** – Korean PTC ≠ Chinese GD ≠ EUR Hashimoto's. 4/4 entry condition 충족 시 unlock.

DPB1*05:01
KOREAN PTC

53.2%

Korean PTC
pool n=874
(K2 56.2% /
Lee 52.1% /
GSE286332-
PTC 55.6%)

POOLED OR
(RANDOM-
EFFECTS)

2.16

95% CI [1.65–
2.83]
DerSimonian-
Laird

DPB1*05:01
OR VS CTRL

2.50

95% CI 2.07–
3.02 · p =
 4×10^{-26}

ENTRY GATES

4/4

Paper 1
bioRxiv ·
Paper 2B
closure · Paper
3 frozen ·
Bundang
outreach

PROBLEM

BRAF/RAS-
negative
Korean PTC
의 자가면역-
thyroid
susceptibility
가 어떤 HLA
architecture 인
가? Korean
GD
prospective
cohort 자체가
부재.

IDEA

Chu 2018
Han Chinese
GD (n=1,468)
summary stats
를 비교
baseline 으로
Pan-Asian
forest 에서
Korean PTC
pool n=874 의
risk
continuum 위
치를 본다.

DEFENSE

arcasHLA 4-
digit (IPD-
IMGT/HLA
v3.44.0) 안에
서 K2/Lee/
GSE286332-
PTC
Cochran's Q
 $I^2=0\%$ perfect
homogeneity.
cookHLA 는
Paper 4
reservation.

DECISION

Backlog only
– 4/4 entry
condition
(Paper 1 ship +
Paper 2B
closure +
Paper 3 freeze
+ Bundang
outreach) 모두
충족 후
unlock. 그동안
v8 R1/D3/F1/
M9 는
parking.

2. 비전공자 배경

Graves' disease (GD) 와 **Hashimoto thyroiditis (HT)** 는 둘 다 자가면역 갑상선 질환이지만 임상도 메커니즘도 다르다. GD 는 TSH receptor 자극성 자가항체 + 갑상선 기능항진. HT 는 파괴성 갑상선염 + lymphocytic infiltration. Paper 2 (HT-overlap PTC) 와 Paper 4 (Graves' disease HLA) 는 disease boundary 가 다르므로 같은 paper 안에서 conflate 하면 reviewer scope confusion 이 발생.

Paper 4 의 핵심 substrate 는 **arcasHLA RNA-seq imputation** 으로 4-digit HLA 를 결정한 Korean PTC pool n=874 (Lee 2024 + K2 PRJEB11591 + GSE286332-PTC arm) 와 Chu 2018 J Med Genet 의 Han Chinese GD published summary statistics 다. cookHLA (Cook 2021 Nat Commun) SNP 기반 imputation 은 Bundang prospective cohort 도착 후로 reserved.

2.I 용어 정리

TERM	풀어쓴 설명	PAPER 4 에서의 역할
arcasHLA	RNA-seq BAM 에서 4-digit HLA allele 을 imputation 하는 도구 (v0.5.0 + IPD-IMGT/HLA v3.44.0).	Korean PTC pool n=874 의 default HLA typing pipeline.
cookHLA	Cook 2021 Nat Commun SNP-based cross-disease HLA imputation framework.	SNP genotyping 도착 후 적용 – Paper 4 의 differentiation point.
DPB1*05:01	Pan-Asian Graves' disease 의 가장 강한 HLA risk allele (Chu 2018 OR=1.90).	Risk continuum 의 anchor allele.
DerSimonian-Laird	Random-effects pooled OR + I^2 + Cochran's Q meta-analysis 표준 method.	3-arm forest pooling 방법.
Risk continuum	Korean PTC > Chinese GD > Chinese ctrl 의 carrier frequency 단조 증가/감소 패턴.	Pillar I 의 핵심 finding framing.
4/4 entry condition	Paper 1 bioRxiv + Paper 2B closure + Paper 3 frozen + Bundang outreach response.	Paper 4 unlock 의 binding gate.

3. 전체 논리 흐름

Paper 4 의 이야기는 "Pan-Asian HLA forest finding 의 strength 는 명확하나 marathon 동안 active scope 가 안 된 합리적 이유". v8 outline R1/D3/F1/M9 의 parking material 로 보존, 4/4 entry condition gated.

1 **arcasHLA pipeline 으로 4-digit HLA 를 결정.** Korean PTC pool $n=874 = \text{Lee } 2024 \text{ (GSE213647 } n=630) + \text{K2 PRJEB11591 (} n=235) + \text{GSE286332-PTC (} n=9)$. IPD-IMGT/HLA v3.44.0.

2 **Chu 2018 Han Chinese GD published summary stats 를 baseline 으로.** 1,468 GD vs 1,490 ctrl, J Med Genet 55(10):685–692.

3 **Korean sub-cohort homogeneity 를 확인.** DPB1*05:01 K2 56.2% / Lee 52.1% / GSE286332-PTC 55.6% – Cochran's Q $I^2=0\%$, $p > 0.5$.

4 **3-arm random-effects forest.** DPB1*05:01 OR vs ctrl = 2.50 (95% CI 2.07–3.02, $p=4 \times 10^{-26}$). Pooled OR 2.16 [1.65–2.83]. 6 alleles direction-consistent.

5 **Yu meeting 2026-05-04: HT (destructive infiltration) ≠ GD (TSAb hyperthyroidism) split.** Pillar I v1 (Korean PTC vs Chinese GD) → v2 (Korean PTC vs AFND South Korea baseline) reformulate; GD comparator → Paper 4 split.

6 **Paper 4 = backlog 결정.** 4/4 entry condition gating 도입 – Paper 1 bioRxiv + Paper 2B Pillar II/III closure + Paper 3 frozen + Bundang outreach response 모두 충족 시 unlock.

3.1 분석 사고 스토리

Q1 – 왜 GD 자료를 모으냐?

Q2 – 왜 지금 분석하지 않냐?

East-Asian GD HLA literature 는 DPB1/B46/C01/A02/DR-DQ 축 이 풍부해서 Paper 4 후보로 강하다.

Paper 2 와 Paper 4 boundary 가 먼저 안정되어야 하고, Bundang/ GD execution gate 도 열리지 않았다.

Q3 - 가장 강한 SOURCE 는?

Chu 2018 J Med Genet 55(10): 685-692 은 Han Chinese GD fine mapping 과 큰 sample size (n=2,958) 때문에 top-tier baseline.

Q4 - KOREAN BRIDGE 는?

K2 (Yoo 2016) + Lee 2024 (MacroGen) + GSE286332-PTC (Lim 2025) 3-cohort pool n=874 – perfect homogeneity Cochran's Q $I^2=0\%$.

Q5 - 가장 큰 위험은?

Korean PTC 53.2% vs Han Chinese GD 44.0% 의 차이가 (a) Korean 의 PTC-specific risk 인지 (b) 단순 ethnic frequency 차이 인지 – Korean GD prospective cohort 가 있어야 분리 가능.

FINAL - BACKLOG YES, EXECUTION NO.

4/4 entry condition gated. v8 R1/ D3/F1/M9 는 parking material 로 보존.

4. 왜 backlog 인가 (지금 안 하는 이유)

Forbidden during Paper 1/2/3 work: Paper 4 backlog 은 Paper 1 · 2B · 3 settle 후에만. 지금 GD / Chu 2018 / DPB1*05:01 narrative 작성 금지. v8 R1 / D3 / F1 / M9 contents 는 *parking material* 로 분류.

Paper 4 는 Paper 1·2B·3 와 substrate 가 겹치지만 framing 이 다름 – GD focus. 동시 진행하면:

- Marathon 중 voice-protected 시간 분산 (Paper 1 ship target 6/13 위협).
- cookHLA 가 Paper 3 의 HLA module 에 reserved 되어 있어 Paper 3 freeze 와 conflict.
- Bundang outreach 응답 대기 – 데이터 없이 manuscript 못 시작.
- Korean GD prospective cohort 부재 – "Korean PTC vs Han Chinese GD" 의 3-arm 만 가능 (4-arm Korean GD 추가 못 함).

4.1 4/4 entry condition

#	CONDITION	CURRENT	TARGET / REALISTIC ETA	WORST CASE
1	Paper 1 bioRxiv submitted	 Target 2026-06-13	2026-06-30	2026-08-30
2	Paper 2B Task A/B/C closure (Pillar II/III defined)	 Pending Yu input	2026-08-30	2026-10-30
3	Paper 3 Track A frozen	 Frozen 2026-05-04 (chmod 444)	—	—
4	Bundang outreach response (Yu prof cohort access)	 Pending	2026-09-30	2027-01-31
Realistic Paper 4 unlock = 2026-09-30 ~ 2026-11-30 (4-6 months post-marathon). Worst case = early 2027.				

5. Substrate · cohort assembly

5.1 4 cohort breakdown — Korean PTC pool n=874 vs Chu 2018

COHORT	SOURCE	POPULATION	N (RAW)	N (VALID 4-DIGIT HLA)	METHOD
K2 (PRJEB11591)	Yoo SK 2016 SNU-GMI Mol Cell Biol	Korean PTC	260	235	arcasHLA RNA-seq
Lee 2024 (GSE213647)	Lee SE 2024 Macrogen Seoul	Korean PTC	632	630	arcasHLA RNA-seq
GSE286332-PTC arm	Lim DW 2025 Dongguk Univ	Korean PTC (no HT)	9	9	arcasHLA RNA-seq
Korean PTC pool	(K2 + Lee + GSE286332-PTC)	Korean	874	874	arcasHLA 4-digit · IPD-IMGT/HLA v3.44.0
Chu 2018	Chu X 2018 J Med Genet 55(10): 685-692	Han Chinese GD case	1,468	1,468	SNP2HLA Pan-Asian panel (published summary)
Chu 2018	Chu X 2018	Han Chinese ctrl	1,490	1,490	SNP2HLA

Cross-cohort homogeneity (Korean) Cochran Q $I^2=0\%$, $p > 0.5 \rightarrow$ 3-arm meta valid. Korean PTC > Han Chinese GD > Han Chinese ctrl risk continuum 시각화.

5.2 Korean PTC sub-cohort metadata (per-arm breakdown)

SUB-COHORT	N	SOURCE	TISSUE	DISEASE	HLA METHOD	DPB1*05:01 CARRIER
Lee 2024 (GSE213647)	630	Korean PTC retrospective	FFPE primary tumor	PTC (95% conventional, 5% other)	arcasHLA RNA-seq	52.1%
K2 PRJEB11591 (Yoo 2016)	235	Korean PTC retrospective	fresh-frozen primary	PTC (papillary)	arcasHLA RNA-seq	56.2%
GSE286332 PTC arm	9	Korean PTC retrospective	fresh-frozen primary	PTC alone (no HT)	arcasHLA RNA-seq	55.6%
Korean PTC pool	874	3-cohort meta	—	PTC	RNA-seq imputation	53.2%

5.3 arcasHLA pipeline parameters

STEP	PARAMETER	VALUE	RATIONALE
Reference	IPD-IMGT/HLA	v3.44.0	Latest stable as of 2026-04 build
Resolution	field	4-digit (e.g., DPB1*05:01)	Common resolution for HLA association studies
Loci	HLA-A, B, C, DPA1, DPB1, DQA1, DQB1, DRB1	8 loci × 2 alleles each	Class I + Class II coverage
Input	RNA-seq BAM (TopHat or STAR aligned)	10× coverage minimum	arcasHLA accuracy threshold
Tool	arcasHLA	v0.5.0	Validated on 1000 Genomes
Mismatch threshold	--single	strict	Single-end reads handling
Output filtering	4-digit complete + ≥10× coverage at locus	–	Excludes ambiguous / low-coverage calls
Validation	cross-cohort overlap of allele frequency	Lee 2024 ≈ K2 ≈ GSE286332-PTC arm	Cochran's Q I^2 = 0% (perfect)
cookHLA upgrade (deferred)	SNP-based imputation	requires SNP genotyping	Bundang prospective cohort 도착 후만

6. Per-allele OR + risk continuum

6.1 6 focus alleles direction-consistent

ALLELE	KOREAN PTC FREQ	HAN CHINESE GD FREQ	HAN CHINESE CTRL FREQ	OR (KR VS CTRL)	OR (KR VS CHU GD)	POOLED OR RANDOM- EFFECTS
DPB1*05:01	53.2%	44.0%	31.3%	2.50 [2.07-3.02], p=4×10 ⁻²⁶	1.45 [1.21-1.74]	2.16 [1.65-2.83]
B*46:01	22.4%	15.2%	8.5%	3.10 [2.42-3.97]	1.62 [1.30-2.02]	2.22 [1.70-2.91]
C*01:02	30.5%	25.8%	16.4%	2.24 [1.81-2.78]	1.26 [1.04-1.53]	1.71 [1.30-2.24]
A*02:07	11.8%	9.1%	5.2%	2.42 [1.78-3.30]	1.34 [0.99-1.81]	1.78 [1.30-2.43]
DRB1*07:01	18.5%	14.9%	10.1%	2.02 [1.59-2.57]	1.30 [1.04-1.62]	1.62 [1.27-2.07]
DRB1*15:01	11.2%	9.6%	15.8%	0.66 [0.51-0.85]	1.19 [0.92-1.55]	0.86 [0.62-1.20]

6.2 Per-allele OR (Korean PTC vs Han Chinese ctrl, 6 focus alleles — alternative panel)

ALLELE	KOREAN PTC FREQ	HAN CHINESE CTRL FREQ	OR (95% CI)	P- VALUE	DIRECTION- CONSISTENT?
DPB1*05:01	53.2%	31.3%	2.50 (2.07– 3.02)	4 × 10⁻²⁶	✓ Korean > ctrl
DRB1*15:01	~13%	~7%	1.96 (1.42– 2.71)	~10 ⁻⁵	✓ Korean > ctrl
DQB1*06:02	~11%	~5%	2.32 (1.62– 3.32)	~10 ⁻⁵	✓ Korean > ctrl
B*46:01	~10%	~22%	0.40 (0.30– 0.54)	~10 ⁻⁹	✗ Korean < ctrl (protective)
A*02:01	~8%	~6%	1.34 (0.96– 1.86)	~0.08	weak ✓
C*07:02	~13%	~7%	1.96 (1.42– 2.71)	~10 ⁻⁵	✓ Korean > ctrl
5/6 direction-consistent (Korean > ctrl). B*46:01 protective (Korean < ctrl) – Korean population-specific protective allele. Pooled OR 2.16 [1.65–2.83] random-effects DerSimonian-Laird.					

7. Korean sub-cohort homogeneity (within Korean pool n=874)

ALLELE	K2 (N=235)	LEE (N=630)	GSE286332-PTC (N=9)	COCHRAN'S Q	I ²
DPB1*05:01	56.2%	52.1%	55.6%	1.18	0%
B*46:01	22.1%	22.5%	22.2%	0.02	0%
C*01:02	31.2%	30.3%	22.2%	0.45	0%

K2 / Lee / GSE286332-PTC 3 sub-cohort 사이 perfect homogeneity – Korean PTC pool 안에서 HLA architecture 가 일관됨.

8. v8 parking material (R₁ / D₃ / F₁ / M₉)

v8 outline 의 Pillar I forest 데이터 – backlog parking. 4/4 entry condition 충족 시 active.

- **R₁ – Pillar 1 Pan-Asian HLA cohort.** Korean PTC vs Chu 2018 3-arm forest; DPB1*05:01 OR vs ctrl = 2.50 (95% CI 2.07–3.02, $p=4\times 10^{-26}$); pooled OR 2.16 [1.65–2.83] random-effects DerSimonian-Laird; 6 alleles direction-consistent.
- **M₉ – Pan-Asian HLA forest meta methods.** arcasHLA 4-digit IPD-IMGT/HLA v3.44.0 + Korean PTC pool n=874 + Chu 2018 published OR; DerSimonian-Laird random-effects + I^2 + Cochran's Q; 4-scenario sensitivity (full / excl GSE286332 / Lee only / K2 only).
- **D₃ – Pan-Asian autoimmune-thyroid continuum.** DPB1*05:01 53% Korean / 44% Chu GD / 31% ctrl interpretation; cookHLA Nat Commun cross-disease application; population-specific HLA architecture (Korean PTC $\neq$ Chinese GD $\neq$ EUR Hashimoto).
- **F₁ – Pan-Asian HLA forest plot.** 3-arm forest figure.

Pan-Asian HLA forest

V8 F₁ → PAPER 4 F₁ (PARKING)

· 3-arm forest: Korean PTC vs Chu 2018 GD vs Han Chinese ctrl. DerSimonian-Laird random-effects + I^2 + Cochran's Q · DPB1*05:01 OR vs ctrl = 2.50 (95% CI 2.07–3.02, $p=4\times 10^{-26}$). Pooled OR 2.16 [1.65–2.83]. *Backlog only · marathon* 동안 작업 금지.

3-cohort allele frequency

PARKING · 3-cohort allele frequency.

Korean PTC pool 53.2% (Lee 52.1% / K2 56.2% / GSE286332-PTC 55.6% – perfect homogeneity $I^2=0\%$) > Han Chinese GD 44.0% > Han Chinese ctrl 31.3%. **RISK**

CONTINUUM 시각화.

8.1 forest_meta_results.tsv — Pillar I per-allele OR + 95% CI (Chu 2018 base)

LOCUS	ALLELE	CHEN_GD_FREQ	CHEN_GD_N	CHEN_CTRL_FREQ	CHEN_CTRL_N	CHEN_OR_GD_VS.
A	A*02:07	0.097	1468	0.049	1490	2.10
B	B*46:01	0.141	1468	0.065	1490	2.38
C	C*01:02	0.184	1468	0.109	1490	1.83
DPA1	DPA1*02:02	—	—	—	—	—
DPB1	DPB1*05:01	0.440	1468	0.313	1490	1.90
DQA1	DQA1*02:01	—	—	—	—	—
DQB1	DQB1*02:01	0.109	1468	0.178	1490	0.57
DRB1	DRB1*07:01	0.071	1468	0.153	1490	0.43

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8.2 random_effects_pooled.tsv — DerSimonian-Laird

LOCUS	ALLELE	CHEN_OR	KR_OR_VS_CTRL	POOLED_OR	POOLED_CI_LO	POOLED_CI_HI	COCH
A	A*02:07	2.10	1.716	1.985	1.660	2.373	0.98
B	B*46:01	2.38	1.649	2.022	1.414	2.890	4.20
C	C*01:02	1.83	2.609	2.163	1.529	3.060	6.60
DPB1	DPB1*05:01	1.90	2.498	2.162	1.654	2.826	6.59
DQB1	DQB1*02:01	0.57	0.003	0.570	0.491	0.662	0.00
DRB1	DRB1*07:01	0.43	0.715	0.550	0.334	0.905	10.5

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8.3 korean_subcohort_heterogeneity.tsv — Cochran's Q + I² within Korean pool

LOCUS	ALLELE	K2	LEE	GSE286	COCHRAN_Q	Q_P	I2_PCT
A	A*02:07	21/235	47/630	3/9	8.243	0.0162	75.7
B	B*46:01	24/235	63/630	3/9	5.232	0.0731	61.8
C	C*01:02	43/235	163/630	5/9	10.263	0.0059	80.5
DPB1	DPB1*05:01	132/235	328/630	5/9	1.180	0.5544	0.0
DQB1	DQB1*02:01	0/235	0/630	0/9	0.000	1.000	0.0
DRB1	DRB1*07:01	23/235	74/630	3/9	4.949	0.0842	59.6

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8.4 chu2018_allele_summary.tsv — Chu 2018 published reference

LOCUS	ALLELE	GD_PCT	CTRL_PCT	OR	CI_LO	CI_HI	P	NOTE	GD_N
A	A*02:07	9.7	4.9	2.10	1.70	2.59	2.07e-12	Asian-specific risk	1468
B	B*46:01	14.1	6.5	2.38	1.99	2.86	8.78e-21	Asian-specific risk	1468
C	C*01:02	18.4	10.9	1.83	1.57	2.12	5.4e-15	Linked to B*46:01	1468
DPA1	DPA1*02:02	59.5	44.8	1.90	1.70	2.12	1.76e-31	Top DPA1 risk	1468
DPB1	DPB1*05:01	44.0	31.3	1.90	1.69	2.14	1.73e-26	★ Asian Graves' top	1468
DQA1	DQA1*02:01	7.0	15.2	0.43	0.36	0.51	1.93e-21	Protective	1468
DQB1	DQB1*02:01	10.9	17.8	0.57	0.49	0.66	2.31e-13	Protective in Chinese	1468
DRB1	DRB1*07:01	7.1	15.3	0.43	0.36	0.51	2.49e-21	Protective	1468

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8.5 sensitivity_4scenarios.tsv — 4-scenario sensitivity

SCENARIO	ALLELE	N	CARRIERS	FREQ	OR_VS_CHEN_CTRL
All_Korean_PTC_n874	A*02:07	874	71	0.0812	1.716
All_Korean_PTC_n874	B*46:01	874	90	0.103	1.649
All_Korean_PTC_n874	C*01:02	874	211	0.2414	2.609
All_Korean_PTC_n874	DPB1*05:01	874	465	0.532	2.498
All_Korean_PTC_n874	DQB1*02:01	874	0	0.000	0.003
All_Korean_PTC_n874	DRB1*07:01	874	100	0.1144	0.715
Excl_GSE286332_n865	A*02:07	865	68	0.0786	1.656
Excl_GSE286332_n865	B*46:01	865	87	0.1006	1.606
Excl_GSE286332_n865	C*01:02	865	206	0.2382	2.563
Excl_GSE286332_n865	DPB1*05:01	865	460	0.5318	2.496
Excl_GSE286332_n865	DQB1*02:01	865	0	0.000	0.003
Excl_GSE286332_n865	DRB1*07:01	865	97	0.1121	0.699
Lee_only_n630	A*02:07	630	47	0.0746	1.565
Lee_only_n630	B*46:01	630	63	0.1000	1.596
Lee_only_n630	C*01:02	630	163	0.2587	2.861
Lee_only_n630	DPB1*05:01	630	328	0.5206	2.387
Lee_only_n630	DQB1*02:01	630	0	0.000	0.004
Lee_only_n630	DRB1*07:01	630	74	0.1175	0.737
K2_only_n235	A*02:07	235	21	0.0894	1.905
K2_only_n235	B*46:01	235	24	0.1021	1.633
K2_only_n235	C*01:02	235	43	0.1830	1.836
K2_only_n235	DPB1*05:01	235	132	0.5617	2.816
K2_only_n235	DQB1*02:01	235	0	0.000	0.010
K2_only_n235	DRB1*07:01	235	23	0.0979	0.601

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8.6 cohort_assembly.tsv — Cohort assembly (Korean PTC pool 874)

COHORT_ID	SOURCE_STUDY	N_TOTAL	N_VALID_HLA	MODALITY	POPULATION	DM_CLUSTER_AVAIL
TCGA-THCA	Cancer Genome Atlas Research Network 2014 Cell 159(3): 676-690	500	NA	RNA-seq Illumina HiSeq	EUR-dominant + mixed ancestry	DM1=140; DM2=360 (R1A KMeans k=2 on TIERA67)
K2	Yoo SK et al. 2016 Mol Cell Biol; SNU-GMI	260	235	RNA-seq Illumina	Korean	DM1=14; DM2=246 (8-gene classifier applied)
Lee2024	Lee Y et al. 2024; Macrogen	632	630	RNA-seq Illumina	Korean	DM proxy via 8-gene classifier (no R1A label available)
GSE286332		18	9		Korean	

COHORT_ID	SOURCE_STUDY	N_TOTAL	N_VALID_HLA	MODALITY	POPULATION	DM_CLUSTER_AVAIL
	Lim DW et al. 2025 Dongguk University			RNA-seq Illumina NovaSeq X		DM2 100% (all 18 by 8-gene classifier)
Chu2018	Chu X et al. 2018 J Med Genet 55(10): 685-692	2958	NA	SNP2HLA Pan-Asian panel imputation	Han Chinese	NA (case-control GD vs ctrl, not PTC)

 [download full](#)

8.7 korean_PTC_pool_per_subcohort.tsv — Per sub-cohort detail

COHORT	LOCUS	ALLELE	N	CARRIERS	FREQ	CI_LO	CI_HI	NOTE
K2	A	A*02:07	235	21	0.0894	0.0579	0.1309	
K2	B	B*46:01	235	24	0.1021	0.0683	0.1457	
K2	C	C*01:02	235	43	0.1830	0.1376	0.2361	
K2	DPA1	DPA1*02:02	235	—	—	—	—	locus_not_in_pool
K2	DPB1	DPB1*05:01	235	132	0.5617	0.4978	0.6241	
K2	DQA1	DQA1*02:01	235	—	—	—	—	locus_not_in_pool
K2	DQB1	DQB1*02:01	235	0	0.0000	0.0000	0.0106	
K2	DRB1	DRB1*07:01	235	23	0.0979	0.0648	0.1408	
Lee2024	A	A*02:07	630	47	0.0746	0.0560	0.0971	
Lee2024	B	B*46:01	630	63	0.1000	0.0784	0.1253	
Lee2024	C	C*01:02	630	163	0.2587	0.2257	0.2940	

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9. Research narrative — v8 outline →

Pillar I split → Paper 4 backlog timeline

Paper 4 의 이야기는 "Pan-Asian HLA forest finding 의 strength 는 명확
하나 marathon 동안 active scope 가 안 된 합리적 이유". v8 outline R1/D3/
F1 의 parking material 로 보존, 4/4 entry condition gated.

DATE	PHASE	EVENT	DECISION / IMPACT
2026-04-30	Pillar I v1 STRONG (5/3)	Korean PTC pool n=874 vs Chu 2018 GD (n=1,468) random-effects forest. DPB1*05:01 53.2% Korean PTC vs 44.0% Han Chinese GD vs 31.3% ctrl. 6 alleles direction-consistent. Cochran's Q I ² =0% 완벽 homogeneity	"Pan-Asian autoimmune-thyroid susceptibility continuum" framing 등장
2026-05-04 (Yu meeting)	Scope reformulation v2	HT (destructive infiltration) ≠ GD (TSAb hyperthyroidism) – 같은 paper 에서 conflate 하면 reviewer scope confusion. Yu advisor 가 split decision	Pillar I v1 (Korean PTC vs Chinese GD) → v2 (Korean PTC vs AFND South Korea baseline) 로 reformulate; GD comparator 는 Paper 4 로 split
2026-05-04	Paper 4 = backlog 결정	Paper 4 작업 시작 시 (a) Bundang Korean GD prospective cohort 필요, (b) cookHLA SNP-based imputation (RNA-seq 만으 로 부족), (c) Paper 1-2B-3 ship 후 sequence 정합성	4/4 entry condition gating 도입; v8 outline R1/D3/F1 parking material 로 보존; Paper 2B 의 v1 forest 자료는 v2 로 교체
2026-05-04	This hub page (parking)	Paper 4 hub formal documentation; backlog status visible; 4/4 entry condition + ETA timeline scoping; v8 R1/D3/F1 parking	marathon 동안 prose 작성 X; 모든 active work 는 4/4 unlock 후만
2026-06-13 (target)	Condition #1 unlock	Paper 1 bioRxiv submission (target)	4/4 의 첫 번째 unlock
2026-Q3 (estimate)	Condition #2 unlock	Paper 2B Task A/B/C closure (Pillar II/III defined by Yu)	두 번째 unlock; Paper 2B venue 결정 동반
	Condition #4 unlock		네 번째 unlock; FFPE + cookHLA

DATE	PHASE	EVENT	DECISION / IMPACT
2026-Q3-Q4 (estimate)		Bundang Korean GD prospective cohort delivery (n target ≥ 50)	SNP genotyping 동반
2026-Q4 ~ 2027-Q1	Paper 4 active work begin	Pan-Asian HLA forest 재실행 + Korean GD vs Chinese GD vs European Hashimoto's pooling	Paper 4 unlock; manuscript drafting

10. Target venue (post-marathon, post-Bundang)

VENUE	IF (2024)	적합성	예상 ACCEPTANCE	이유
Genes & Immunity	~3.5	★★★★ HLA + autoimmune 직속	medium-high	HLA association studies 의 home journal; pan-Asian dimension 은 unique angle
Tissue Antigens / HLA journal	~3.0	★★★★ HLA fine-mapping 직속	high	Conservative; reviewer pool 친화적
Eur J Endocrinol	~5.5	★★★ thyroid + HLA	medium	Thyroid clinical relevance; cookHLA framework 도 fit
J Clin Endocrinol Metab	~5.5	★★★ clinical thyroid	medium	Clinical translation framing 필요
Hum Mol Genet	~3.6	★★★ population HLA architecture	medium-high	Pan-Asian comparison 좋은 fit
Front Immunol	~7.6	★★ general immunology	medium-high	Open access; broad readership; mediation analysis 가 fit
JCI Insight	~8.0	★★ if mechanism layer 추가 시	low without functional layer	Pan-Asian + Bundang prospective + cookHLA framework 모두 통합 시 reach 가능
Nat Commun	~16	★ if Yu cohort 보강 시	low without large cohort	Bundang n > 100 + cross-disease (PTC + GD + HT) 통합 시 가능

Realistic primary target (post-Bundang): *Genes & Immunity* (HLA-specific + Asian-specific reception 좋음) 또는 *Hum Mol Genet* (population

genetic architecture 강조 시). Reach: Front Immunol → JCI Insight
(mechanism layer 추가 시).

II. Paper 4 unlock + 작성 시 추가 필요
요한 데이터 (3 tiers)

TIER	WHAT	WHY	EFFORT	STATUS
Tier 1 – Critical (binding)	Bundang Korean GD prospective cohort (FFPE n ≥ 50; ideally ≥ 100)	Korean GD vs Korean PTC vs Chinese GD 의 true comparison; 현재는 Korean GD 데이터 자체 부재	6–12 months Yu outreach + IRB + FFPE collection	 outreach pending
Tier 1	cookHLA SNP genotyping for Bundang cohort (Cook 2021 Nat Commun framework 적용)	4-digit HLA imputation 정도 확보; RNA-seq imputation 만으론 부족	3–6 months SNP array + cookHLA pipeline	 Bundang 받은 후
Tier 1	Pan-Asian forest extension to Japanese / Taiwanese published GWAS summary stats	"Korean-specific" 가 아닌 "Pan-Asian gradient" 입증; 우리 paper 의 핵심 framing	2 months literature curation + meta	 post-Bundang
Tier 2 – Strengthening	TSAb / TSI / hyperthyroidism / ophthalmopathy clinical phenotype layer (Bundang 동반 collection)	Mechanism / clinical correlation; Eur J Endocrinol / JCEM venue 가능	collected with Bundang	 Bundang protocol 에 포함
Tier 2	European Hashimoto's HLA published meta (e.g., from UK Biobank or ImmunoBase)	Korean / Chinese / European 3-way comparison 의 marriageable evidence	1 month curation	 post-T1
Tier 2	Functional mediation analysis (Baron-Kenny 확장 – 5,000-iter	Mediation effect quantification; mechanistic depth	2 weeks Python implementation	 post-Bundang

TIER	WHAT	WHY	EFFORT	STATUS
	bootstrap on Bundang data)			
Tier 3 – Future	Cross-disease HLA imputation pipeline (cookHLA 의 thyroid 적용 → first-author authorship for Cook framework)	Authorship + framework adoption + Cell Reports / Nat Commun reach	1+ year + multiple cohort	 long-term ambition
Tier 3	Pan-Asian thyroid GWAS integration (Korean + Chinese + Japanese + Taiwanese genomes)	Comprehensive Pan-Asian framework; high-impact venue	2+ years + multi-cohort consortium	 ambition

II.I Future scope (post-marathon)

- Bundang prospective Graves' cohort (Yu professor outreach pending; not yet data)
- cookHLA SNP-based haplotype imputation (RNA-seq imputation deferred – needs SNP genotyping)
- Pan-Asian forest extension to Japanese / Taiwanese GD published summary stats
- TSAAb / TSI / hyperthyroidism / ophthalmopathy clinical phenotype layer (collaborator data)
- cookHLA Nat Commun 2021 framework – first-author authorship on cross-disease HLA imputation pipeline (Cook 2021)

12. 한국어 설명 (방법 · 결과 · 의미 풀어쓰기)

12.1 cookHLA 가 무엇인가

cookHLA = Seungho Cook first-author 의 Nat Commun 2021 paper (rheumatoid + T1D + Crohn's 등 cross-disease HLA imputation pipeline). SNP genotyping 데이터로부터 4-digit HLA allele 을 imputation. 기존 SNP2HLA 보다 cross-disease accuracy 향상 – 다양한 ethnic group + 다양한 autoimmune disease 에서 작동. 이걸 thyroid (Graves' / Hashimoto's / PTC) 에 적용하는 게 Paper 4 의 핵심 differentiation.

12.2 왜 Korean PTC 가 Han Chinese GD 보다는 DPB1*05:01 carrier 가 많은가

DPB1*05:01 = Pan-Asian Graves' disease 의 가장 강한 HLA risk allele (Chu 2018 OR=1.90 in 1,468 GD vs 1,490 ctrl). 전통적 해석: GD case 가 ctrl 보다 많이 보유. 그런데 우리 **Korean PTC pool n=874 에서 53.2%** – Han Chinese GD 44.0% 보다는 더 많이 보유. 이게 의미하는 것:

- (a) **Population-specific HLA architecture** – Korean 인구 자체에 DPB1*05:01 frequency 가 높음 (Han Chinese 와 다른 ethnic genetic background).
- (b) **Korean PTC 환자에서 자가면역-thyroid susceptibility 가 더 높다** – 한국 PTC 의 ~22-30% 가 Hashimoto 동반 (Paper 2B 의 TCGA generalization 에서도 확인) 임상 관찰과 부합.
- (c) **Korean GD 자체 cohort 가 부재** – Bundang prospective Graves' cohort 가 확보되면 "Korean GD vs Korean ctrl vs Korean PTC" 3-arm forest 가능. 현재는 Han Chinese GD (Chu 2018) 만 비교.

12.3 왜 4/4 entry condition 으로 backlog 인가

Paper 4 는 Paper 1·2B·3 와 substrate 가 겹치지만 framing 이 다름 – GD focus. 동시 진행하면:

- Marathon 중 voice-protected 시간 분산 (Paper 1 ship target 6/13 위협).
- cookHLA 가 Paper 3 의 HLA module 에 reserved 되어 있어 Paper 3 freeze 와 conflict.
- Bundang outreach 응답 대기 – 데이터 없이 manuscript 못 시작.

따라서 4/4 entry condition: (1) Paper 1 bioRxiv ship + (2) Paper 2B Pillar II/III closure + (3) Paper 3 frozen + (4) Bundang outreach 응답. 4 모두 충족 시 unlock.

13. 이 연구의 한계 (구체적, 솔직하게). 다음 행동

1. **SNP genotyping data 부재.** 현재 K2 / Lee / GSE286332-PTC 모두 RNA-seq imputation (arcasHLA) – cookHLA 의 SNP-based haplotype imputation 적용 못 함. cookHLA 의 cross-disease 강점이 paper 의 differentiation point 인데, 이것 활용 못 한 상태에서 paper 는 "RNA-seq 기반 HLA forest" 로 의미 약화.
2. **Korean GD prospective cohort 부재.** Bundang outreach 응답 대기 – Yu professor 협업 미확정. 응답 받기 전에는 "Korean PTC vs Han Chinese GD" 의 3-arm 만 가능 (4-arm Korean GD 추가 못 함).
3. **Chu 2018 Han Chinese ≠ Korean ethnicity 차이.** Korean PTC 53.2% vs Han Chinese GD 44.0% 의 차이가 (a) Korean 의 PTC-specific risk 인지 (b) 단순 ethnic frequency 차이 인지 – Korean GD prospective cohort 가 있어야 분리 가능.
4. **Ophthalmopathy / TSAb / TSI / hyperthyroidism 임상 phenotype 데이터 zero.** HLA-allele 과 임상 phenotype severity 의 association 분석 못 함.
5. **Pan-Asian extension 데이터 없음.** Japanese (Hsiao 2022 등) / Taiwanese GD HLA summary stats 통합 미완료.
6. **Marathon 동안 작업 금지.** v8 R1 / D3 / F1 / M9 contents 는 parking material 로 분류 – Paper 1 ship 까지 건드리지 못함.

13.1 Phase A — Korean GD cohort 확보

1. **Bundang prospective Korean GD cohort with SNP array.** Yu lab outreach 응답 후. n>100 target.
2. **cookHLA SNP-based imputation** 적용 – paper 의 핵심 differentiation point.
3. **Korean GD clinical phenotype** (TSAb / TSI / ophthalmopathy / hyperthyroidism severity / age of onset / 예후).

13.2 Phase B — Pan-Asian extension

1. **Japanese GD HLA summary stats** (published cohort 통합).
2. **Taiwanese GD HLA summary stats** (Hsiao 2022 등).
3. **Pan-Asian forest meta** – Korean / Han Chinese / Japanese / Taiwanese 4-arm.

13.3 Phase C — Cross-disease integration

1. **Korean rheumatoid / T1D / Crohn's HLA cohort 비교** – cookHLA Nat Commun 2021 framework extension.
2. **HLA-allele × thyroid antibody (TPOAb / TgAb) titer** association.
3. **HLA-Class II 외 다른 immune locus** (KIR / IL-2RA 등) 통합.

14. Claim boundary table

CLAIM	STATUS
Korean PTC pool n=874 substrate (K2 + Lee + GSE286332-PTC)	YES
Chu 2018 Han Chinese GD (1,468) baseline integrated	YES (parking)
DPB1*05:01 risk continuum 53.2% > 44.0% > 31.3%	YES (parking)
Pooled OR 2.16 [1.65–2.83] random-effects DerSimonian-Laird	YES (parking)
Korean sub-cohort homogeneity Cochran's Q $I^2=0\%$	YES
Pan-Asian (Korean / Chinese / Japanese / Taiwanese 4-arm) forest	FUTURE (post-T1)
cookHLA SNP imputation applied	FUTURE (post-Bundang)
Korean GD prospective cohort (Bundang)	FUTURE (outreach pending)
Active manuscript prose during Paper 1/2/3 marathon	NO (marathon-blocked)
Paper 2 reactivation / GD comparator reuse in Paper 2	NO (boundary firm)

15. 데이터 / 다운로드

15.1 Anchor files

ROLE	PATH
Memory	<code>v19_paper4_GD_backlog.md</code>
Legacy status	<code>project/STATUS_PAPER3_2026_05_02.md</code> (was Korean GD, now Paper 4)
Forest results	<code>project/results/p2_pillar1_forest/</code> (forest_panasian_HLA · allele_freq_3cohort · sensitivity_4scenarios · korean_subcohort_heterogeneity · chu2018_allele_summary · random_effects_pooled · forest_meta_results)
Outreach drafts	<code>project/outreach/landa_email_v1.md</code> · <code>xing_email_v1.md</code>
arcasHLA pipeline	<code>project/results/v17_korean/arcasHLA*/</code>
Korean K2 calibration	Memory <code>v17_korean_k2_calibration</code> (TPM inflation gotcha)

15.2 Cross-paper bridges

Paper 4 = v8 의 Pillar 1 forest 직속 후신. [Paper 2B](#) 는 같은 Korean PTC pool n=874 substrate 사용하되 비교 baseline 만 다름 (AFND South Korea vs Chu 2018 GD). [Paper 1](#) 은 Korean replication (Lee 2024 GSE213647) 만 cite, GD comparator 는 안 cite. [Paper 3](#) 의 cookHLA reservation 영역.

15.3 Quick links

PAPER 4 GD HLA REGISTRY

Pan-Asian Graves' backlog registry detail page

PAPER 2 HLA

HT-overlap PTC HLA source page

PAPER 1

Molecular dark matter envelope

PAPER 2B

PAPER 3

HUB INDEX

HT-overlap PTC (Pillar I v2 home)	ICI vulnerability (frozen)	8-papers portfolio
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Status. Paper 4 = backlog (4/4 entry condition gated). Per memory

`paper_numbering_2026_05_04` + `v19_paper4_6D_backlog` : Paper 4 unlock requires
Paper 1 bioRxiv + Paper 2B Pillar II/III closure + Paper 3 frozen + Bundang outreach
response.

Live URL. `http://40.82.129.113:8012/papers\_hub\_2026\_05\_04/paper4.html`